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MRI-guided periradicular nerve root infiltration therapy in low-field (0.23-T) MRI system using optical instrument tracking

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Abstract The purpose of this study was to evaluate the feasibility of the MRI-guided periradicular nerve root infiltration therapy. Sixty-seven nerve root infiltrations under MRI guidance were done for 61 patients suffering from lumbosacral radicular pain. Informed consent was acquired from all patients. A 0.23-T open-MRI scanner with interventional tools (Outlook Proview, Philips Medical Systems, MR Technologies, Finland) was used. A surface coil was used in all cases. Nerve root infiltration was performed with MRI-compatible 20-G needle (Chiba type MReye, Cook, Bloomington, Ind.; or Manan type, MD Tech, Florida). The evaluation of clinical outcome was achieved with 6 months of clinical follow-up and questionnaire. The effect of nerve root infiltration to the radicular pain was graded: 1=good to excellent, i.e., no pain or not disturbing pain allowing normal physical activity at 3 months from the procedure; 2=temporary, i.e., temporary relief of pain; 3=no relief of pain; and 4=worsening of pain. As an adjunct to MRI-guided positioning of the needle the correct needle localization by the nerve root was confirmed with saline injection to nerve root channel and single-shot fast spin echo (SSFSE) imaging. The MRI guidance allowed adequate needle positioning in all but 1 case (98.5%). This failure was caused by degeneration-induced changes in anatomy. Of patients, 51.5% had good to excellent effect with regard to radicular pain from the procedure, 22.7% had temporary relief, 21.2% had no effect, and in 4.5% the pain worsened. Our results show that MRI guidance is accurate and safe in performing nerve root infiltration at lumbosacral area. The results of radicular pain relief from nerve root infiltration are comparable to CT or fluoroscopy studies on the subject.

Keywords Interventional MRI · Optical tracking · Low field · Periradicular · Therapy

Introduction

Selective periradicular nerve root infiltration with local corticosteroids and anesthetics has been used for preoperative evaluation of lumbosacral pain and sciatica patients [1, 2], but is has also therapeutic effect in discogenic radicular lumbosacral and sciatic pain [3, 4, 5, 6].

Fluoroscopy and CT have been the gold standard interventional radiological guidance modalities for nerve root infiltration due to good temporal resolution, excellent bone–tissue contrast, and real- or near real-time imaging [7, 8]; however, there is ionizing radiation involved in these modalities. This is troublesome, since these procedures are usually focused on lumbosacral area, are performed predominantly on young individuals still of fertile age, and in many cases multiple therapy sessions are needed thus increasing acquired radiation dose. With CT or fluoroscopy there is also need for local contrast-agent-induced neurogram to verify the correct positioning of the needle [9]. Furthermore, low soft tissue contrast disfavor these methods, although CT has excellent spatial resolution on bone structures. Fluoroscopy guidance in the infiltration of the sacral nerve roots is also technically unsatisfactory [10]. Magnetic resonance imaging in interventional procedures combines the advantages of MRI and enables identification of pain source by local injections [11]. The MRI guidance in various interventional procedures is a topic of intense scientific interest and interventional MRI has already emerged as an option to more conventional methods in selected indications [5, 12, 13]. There are, however, very few studies on MRI-guided therapeutic injections, although MRI has some unique features compared with the other methods, and it has been suggested that MRI provides additional value in performing interventional and diagnostic procedures [14, 15, 16, 17, 18, 19, 20, 21, 22, 23].

The purpose of this study was to evaluate the feasibility and diagnostic accuracy of MRI guidance with optical tracking as a method in performing MRI-guided periradicular nerve root infiltration therapy at lumbosacral area to treat disc-generated pain. An additional goal was to evaluate possible variation in the therapeutic effect to periradicular pain due to different types of disc pathology associated with pain history.

Patients and methods

All consecutive patients referred for MRI-guided percutaneous periradicular nerve root infiltration therapy in our institution during 1 April 1999 to 1 March 2001 were included in the study. Informed consent was obtained from all patients before the procedure. The series consisted of 61 patients who underwent a total of 67 percutaneous periradicular nerve root infiltrations. The periradicular nerve root infiltration was always targeted to the symptomatic side of the patient. Fifty-eight infiltrations were targeted to first sacral root, 7 infiltrations to fifth lumbosacral root, and second to fourth lumbosacral root. Double infiltration was performed in 5 patients over a 4- to 10-month interval from the first procedure due to renewed radicular pain. One patient had periradicular nerve root infiltration also to the contralateral nerve root. The mean age of the patients was 46 years (age range 22–71 years); 41 patients had an unoperated disc disease and 20 patients had history of previous back surgery. All patients had lumbosacral radicular pain or sciatica diagnosis set by surgeon or rehabilitation specialist before referral to the procedure. Patients with indication for acute surgery, such as cauda equina, were excluded from the study. The final outcome of procedural pain relief obtained was evaluated after 6-month clinical follow-up and questionnaire. The effect to the radicular pain was graded as follows:

1. Good to excellent=no pain or not disturbing pain allowing normal physical activity at 3 months from the procedure
2. Temporary=temporary relief of pain (1–4 weeks)
3. No relief of pain
4. Worsening of pain

All the patients had a preprocedural MRI or CT study done.

The possible correlation between the effect of nerve root infiltration therapy and the etiology of the pain was studied. To facilitate this the findings of vertebral disc anatomy and pathology causing possible narrowing in subsequent path of the symptomatic nerve root in individual patients were classified into three categories at the lumbosacral levels from L1 to S1. This classification was done by inspecting the preprocedural imaging data (MRI or CT) obtained from the lumbosacral area. The categories were:

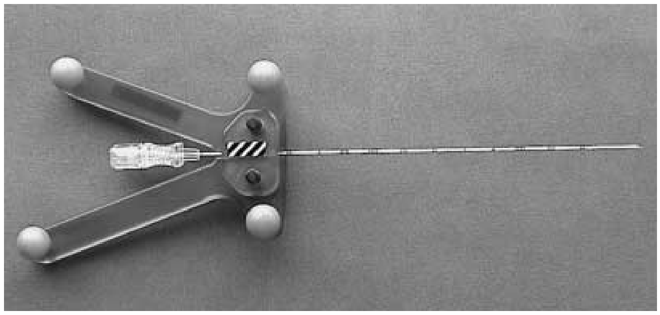

Fig. 1 Needle (Manan type, MD Tech, Florida) and instrument holder for optical tracking

1. Normal or minor protrusion of intervertebral disc without narrowing of the nerve root space
2. Intervertebral disc herniation narrowing the nerve root space
3. Narrowing of the nerve root channel due to post-operational scarring and/or osseal prominence

An MRI-compatible 20-G needle (MReye, Cook, Bloomington, Ind.; or Manan, MD Tech, Florida) was used for nerve root infiltration puncture and injection of therapeutic substance. The MRI scanner used was a 0.23-T open-configuration MRI scanner with interventional optical tracking equipment and software (Outlook Proview, Philips Medical Systems, MR Technologies, Finland). The interventional MRI system consisted of the MRI scanner, in-room monitor, in-room workstation with full scanner control, infrared camera, fixed infrared mirrors, and of foot pedal for operator-controlled sequence start. A surface coil of a single loop type (diameter 20 cm) was used for imaging in all patients during procedure. The tracking was performed with an infrared navigator camera. The camera detects two trackers simultaneously, one of which is attached to the instrument (Fig. 1) and the other to the magnet pole piece, and provides a fixed reference frame. The software supports instrument-guided imaging and the image set center follows the instrument tip. The software calculates the orientation and position of the instrument in a three-dimensional volume. The MR imaging is allowed in a spherical volume of 300 mm in diameter located around the magnet isocenter. Estimate of the tip position and the shaft orientation are based on the assumption of complete rigidness of the calibrated instrument. According to preset values different image sets constructed from different sequences can be obtained and the number of images in set varies from one to nine images.

The tracking software communicates with scanner in real-time via RS-232 serial connection allowing real-time changes in the image plane. The default image planes were set in y-, z-, and x-planes in relation to the instrument tip. The instrument dimensions were calibrated with tracking software before the puncture. The instrument shaft and its trajectory was shown in real-time as a graphic overlay in the image or in the image set during procedure. The guidance software allows to determine the target as a graphic overlay (target point) and gives also spatial feedback from instruments position in relation to the determined target point as a graphic overlay.

A prerequisite for the procedure was a free route to the periradicular nerve root area. The route was determined with T1-weight-

Fig. 2 Setup for the nerve root infiltration. The layout of the surface coil is also shown

ed fast spin-echo (FSE) imaging (5 slices, TE 400 ms, TR 16 ms, slice thickness/interval 7.0 mm/8.0 mm, FOV 380×380 mm, matrix 324×324, acquisition time 23 s), completely balanced steady state (CBASS) imaging (8 slices, CBASS, TR 8.4 ms, TE 4.2 ms, slice thickness/interval 5.0 mm/5.0 mm, FOV 380×380 mm, matrix 256×256, acquisition time 24 s), and T2-weighted FSE images (9 slices, TE 3500 ms, TR 150 ms, slice thickness/interval 7.0 mm/8.0 mm, FOV 380×380 mm, matrix 192×192, acquisition time 35 s) were also obtained whenever no previous MRI imaging had been done. The used sequence, CBASS, is in essence a true fast imaging with steady precession (true-FISP) type sequence combining T2 and T1 expression based on the T2/T1 balance in using high flip angles. This sequence is fast and allows high resolution compared with other fast imaging sequences used in low-field interventional MRI [24].

The nerve root preoperatively determined as most probable pain source by clinical inspection and imaging studies was selected as a target.

Before the operation, all the necessary precautions ensuring normal sterile and safe conditions for a procedure were performed. This included the special safety issues concerning MRI surroundings. The setup for therapeutic MRI-guided injection and the surface coil used are demonstrated in Fig. 2.

The site of skin puncture and procedural route was determined by means of optical tracking. This was done by placing the instrument mounted to tracker close to the skin at the supposed procedure area and taking image sets to obtain information about the procedure area and the route. To reduce procedural subcutaneous pain local anesthetic lidocaine (10%), maximum 5 ml, was then injected into the subcutaneous soft tissue. No other medication, sedation, or monitoring was used. The needle was then advanced through the subcutaneous fat and dorsal sacrolumbar ligaments to the connective tissue surrounding nerve root, i.e., periradicular space using graphic overlay information provided by tracking and instrument-generated susceptibility in repeated fast images as reference (Fig. 3a). The mandrel was removed and 2 ml of saline was introduced into the periradicular space (Fig. 3b). This technique was used in the last 62 infiltrations. The possible pain provocation generated and the nature of it were recorded simultaneously.

Before and during the puncture, fast T1-weighted gradient-echo (3 slices, TR 95 ms, TE 7 ms, slice thickness/interval 7.0 mm/7.0 mm, FOV 380×380 mm, matrix 300×300, acquisition time 12 s; 5 slices, TR 130 ms, TE 11 ms, slice thickness/interval 10.0 mm/10.0 mm, FOV 380×380 mm, matrix 256×256, acquisi-

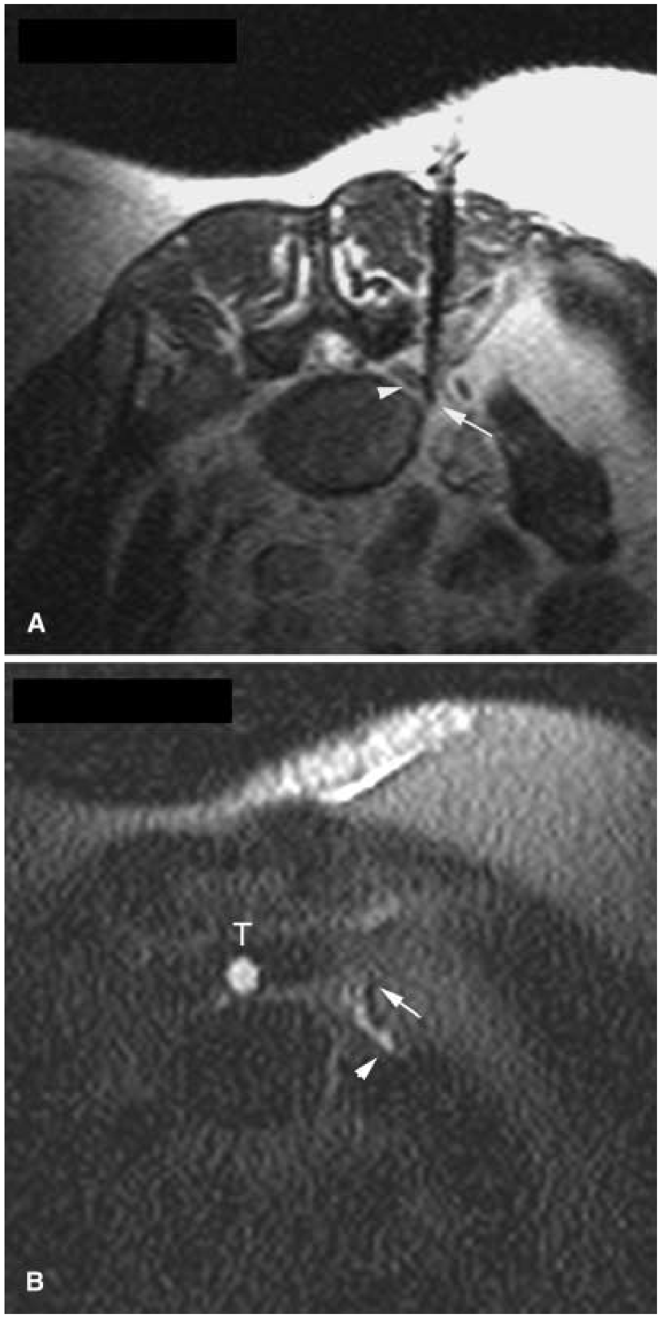

Fig. 3a, b Infiltration of the fifth lumbar nerve root. **a** Final position of the needle. The needle was inserted through the muscle layer and is seen as the black artifact in completely balanced steady state sequence (8 slices, TR 8.4 ms, TE 4.2 ms, slice thickness/interval 5.0 mm/5.0 mm, FOV 380×380 mm, matrix 256×256, acquisition time 24 s). Axial view in the plane of the needle (*arrow* needle tip, *arrowhead* nerve root). **b** Injected saline solution appears bright in the nerve sheath in axial single-shot fast spin echo image (5 slices, TR 274 ms, TE 9000 ms, slice thickness/interval 7 mm/7 mm, FOV 380×380 mm, matrix 256×256, acquisition time 9 s). The projection is the same as in **a** [*arrow* dorsal ramus of the fifth lumbar nerve, *arrowhead* ventral (main) ramus of fifth lumbar nerve]. *T* thecal sac tion time 18 s), or CBASS imaging (8 slices, TR 8.4 ms, TE 4.2 ms, slice thickness/interval 5.0 mm/5.0 mm, FOV 380×380 mm, matrix 256×256, acquisition time 24 s; 1 slice, TR 9.1 ms, TE 4.5 ms, slice thickness/interval 10.0 mm/10.0 mm, FOV 380×380 mm, matrix 256×256 acquisition time 1.5 s) was performed in order to visualize the needle and anatomical structures to enable correct needle positioning to the target.

Immediately before and after the introduction of the saline into the periradicular nerve root space, a single-shot fast spin-echo (SSFSE) imaging (5 slices, TR 9000 ms, TE 274 ms, slice thickness/interval 7 mm/7 mm, FOV 380×380 mm, matrix 256×256, acquisition time 9 s) was performed to detect fluid signal around or parallel to the nerve root. The final proof of correct placement of the needle was obtained with this method. Possible pain provocation by the injection of saline was also used as an adjunct confirmation of the correct placement of the needle.

After this, the therapeutic agent was introduced through the needle, either 2 ml of methylprednisolone–bupivacaine solution (Solomet, methylprednisolone 40 mg/ml–bupivacaine 5 mg/ml, Orion, Finland; 63 infiltrations)or 2 ml bupivacaine (5%; 4 infiltrations) was injected into the periradicular space and the fluid signal around or parallel to the nerve root was again detected with the SSFSE sequence. In the first eight procedures fluoroscopy was used with MRI as an adjunct modality either for determining the puncture site to the skin or to confirm the final placement of the needle in the connective tissue surrounding nerve sheath with a contrast agent.

Table 1 Effect of nerve root infiltration on the radicular pain in all performed punctures

Effect	*N*	%
No pain or not disturbing pain	35	52.2
Temporary relief of pain	15	22.4
No relief of pain	14	20.9
Worsening of pain	3	4.5
Total	67	100.0

Table 2 Effect of nerve root infiltration on the radicular pain according to the operational situation before the procedure

Effect	Patients' operational situation before nerve root infiltration			
	Operated		Not operated	
	N	%	*N*	%
No pain or not disturbing pain	15	78.9	18	41.9
Temporary relief of pain	2	10.5	11	25.6
No relief of pain	2	10.5	11	25.6
Worsening of pain	–	0.0	3	7.0
Total	19	100.0	43	100.0

Results

All the nerve roots selected for targeted infiltration were detected by the low-field MRI device used. The MRI guidance allowed correct needle positioning in all but 1 case (98.5%) when MRI guidance was used. The failure was caused by degeneration-induced changes in anatomy obstructing path to the nerve root canal. There was no need to reposition the needle after initial MRI imaging (CBASS, 8 slices, TR 8.4 ms, TE 4.2 ms, slice thickness/interval 5.0 mm/5.0 mm, FOV 380×380 mm, matrix 256×256, acquisition time 24 s) had confirmed the localization of the needle in the direct vicinity of the nerve root sheath.

The mean puncture time from the introduction of the needle through skin to needle retraction was 12 min, minimum 2 min, and maximum 60 min. The mean procedural time from starting the interventional MRI procedure to completing it was 33 min, minimum 9 min, and maximum 84 min. This included the anesthesia procedures needed. These times did not include the time required for patient and operation room preparation. The mean total number of sequences used in all scans during the procedure was 13, minimum 7 sequences, and maximum 26 sequences. The mean number of used sequences for puncturing was 10, minimum 2, and maximum 20 sequences. There was one case where post-procedural pain was categorized as a post-procedural complication. Three patients underwent back surgery during follow-up period due to persisting clinical symptoms and worsened

Table 3 Therapy effect on patients with different types of nerve root channel narrowing

Effect	Type of finding in preprocedural imaging					
	Disc herniation finding		Narrowing of the nerve root channel findings due to post-operational scarring and/or osseal prominence		Normal or symmetrically protruded disc without narrowing of the nerve root channel findings	
	N	%	*N*	%	*N*	%
No pain or not disturbing pain	23	62.2	2	18.2	3	25.0
Temporary relief of pain	8	21.6	6	54.5	1	8.3
No relief of pain	4	10.8	3	27.3	7	58.3
Worsening of pain	2	5.7	–	0.0	1	8.3
Total	37	100.0	11	100.0	12	100.0

disc pathology findings in imaging studies. All these patients had relief of the pain from surgery.

The therapy effect outcome out of all performed punctures is shown in Table 1.

For this data statistical analysis was performed with a chi-square test where $p<0.001$, $df=3$, and $\chi^2=31.806$.

The effect of nerve root infiltration on the radicular pain according to the operational situation before the procedure is shown in Table 2.

The therapy effect on patients with different types of nerve root channel narrowing is shown in Table 3.

Of all infiltrations, 63 of the infiltrations were done with methylprednisolone–bupivacaine solution, with the outcome of good to excellent in 33 cases (52.4%), temporary relief in 15 cases (23.8%), no effect in 12 cases (19%), and worsening of pain in 3 cases (4.8%). With the four infiltrations done with bupivacaine alone, 2 patients had good to excellent outcome and 2 patients experienced no effect.

Saline solution served as the contrast agent for heavily T2-weighted SSFSE imaging sequences in the last 62 infiltrations; for all of these the contrast-to-noise ratio was sufficient to confirm the correct placement of the needle and allowed visualization of the path of the nerve root treated.

Discussion

In this study we show that MRI-guided nerve root infiltration is safe and accurate. Fluoroscopy or CT-guided procedure is usually the method of choice for nerve root infiltration therapy in the lumbosacral area. Although fluoroscopy is relatively inexpensive and easy to apply, it does have disadvantages of use in guiding interventions. Firstly, the procedural steering must be done according to bony anatomical landmarks and any variation or change in the soft tissue morphology due to anatomical variation or pathology is not detected. The use of contrast media does little to overcome this dilemma; only the edges of adjacent structures to the needle are readily visualized. Secondly, there is radiation burden to the operator and to the patient. Although CT guidance is far better in identifying anatomical structures at the procedure area, it too has restrictions. Despite excellent temporal and spatial resolution, CT is not so good at differentiating various soft tissue structures from each other without special windowing or Hounsfield unit analysis. Computed tomography also shares the problem of radiation with fluoroscopy.

Cross-sectional imaging modalities, such as CT and MRI, are more time-consuming but provide better visualization than fluoroscopy [25]. Computed tomography has been established as a reliable guidance method in demanding interventions [26], but MRI has several features that make it superior to CT as a guidance method. Due to its good sensitivity, MRI is able to detect pathology not seen at all with other modalities. Magnetic resonance imaging also provides superior tissue contrast without contrast medium.

An open-configuration MRI scanner allows good access to the patient. Furthermore, there is seldom any need to move the patient during the procedure. The MRI guidance also allows the freedom to choose any imaging plane. This facilitates the technical task of performing interventional procedures at oblique angles. The optical tracking facilitates real-time control of instrument position in relation to the image data and is easy to use. Combined with the operator-controlled sequence start with foot pedal and fast imaging, the system increased operator control and thus speed and accuracy to the procedure. The advantage of this was also shown in our study, where the initial technical accuracy of almost 100% was achieved almost without complications. The disadvantages of optical tracking are the trackers' need for line of sight and relative insensitivity to needle bending.

The most critical phase of performing minimally invasive procedures is the accurate and safe positioning of the needle. Dissimilar needle artifact sizes in different sequences under MRI guidance must be taken into account [27]. In general, needle artifact in MRI image is much more prominent in T1-weighted FSE and gradient-echo images than in T2-weighted images. The needle thickness affects also the artifact size, and the removal of mandrel diminishes the artifact size in MRI image. Needle position to main magnetic field (B_0) also has considerable effect. The more perpendicular needle position with respect to B_0 orientation produces more pronounced needle artifact. The needle path should be planned accordingly.

The needle bending due to the needle material (titanium alloys) must be taken into account, as the correct position should be ascertained unambiguously. In practice, this means confirmatory images where the needle artifact or injected contrast agent are clearly visible. In MRI-guided nerve root infiltration the needle bending can be easily controlled by thicker slice set if necessary. As an adjunct to optical tracking and direct visualization of the needle we used saline solution as contrast agent to visualize nerve root sheath injected. This proved to be a very effective form of confirming the right placement of the needle. Gadolinium compounds with T1-weighted sequences have been used for this purpose [28]; however, to date, these compounds have not been accepted for intrathecal use and using saline as contrast agent makes it possible to avoid the possible side effects of intrathecal or epidural gadolinium infiltration.

With MRI guidance there is no irradiation burden to the patient or the physician, which is advantageous in performing procedures on adolescents and women potentially pregnant but particularly important for the physician when considering the interventional radiologists' radiation burden generally. It is also generally acknowledged that interventional procedures should be performed without ionizing radiation whenever feasible.

In terms of imaging speed, MRI is comparable to conventional CT but not to CT fluoroscopy. The slower imaging speed in MRI is not a problem if real-time optical tracking is available. Computed tomography has superior axial resolution, and the ability to detect pneumothorax favors the use of CT in the thoracic region; however, in our series where all the action took place in lumbosacral area we found that operative image quality obtained with a 0.23-T scanner was sufficient to guarantee a safe procedure. We had 1 patient where the worsening of the radicular pain was categorized as minor complication due to the puncture. This was probably caused by nerve root irritation from the puncture or either from the therapeutic substances used.

Low back disorders are extremely prevalent in all societies, and the rate of disability as well as the costs have increased, independently of the disease prevalence, over recent decades [29]. It is obvious that we cannot solve this dilemma with increasing surgical interventions. Minimally invasive interventional procedures are an option to relieve pain and minimize the risk of disability.

To our knowledge, this is the first study where usefulness of MR imaging guidance in lumbar nerve root infiltration is evaluated. In the referred studies [3, 4, 10, 30] the outcome of the first sacral root and lumbar root infiltrations have been evaluated in fluoroscopy-, CT-, and MRI guidance. In our study we were able to infiltrate the treatment agent directly into the main, ventral ramus sheath of the lumbosacral nerve roots. Since 1995 over 1400 nerve root infiltrations have been performed with fluoroscopy guidance in our institution, and in our experience this method allows the needle to be placed only into the dorsal root canal or dorsal root sheath of the lumbosacral nerve roots and the depth is difficult to estimate. Magnetic resonance imaging overcomes this problem.

In our study the length of the procedural time decreased gradually from 60 min reaching minimum 2 min. This can be seen as a learning-curve development in implementing the procedure.

Previous reports concerning the effect of nerve root infiltration therapy to the radicular pain have been promising. In a study of CT-guided lumbar nerve root infiltrations [30], 55% of patients were free of symptoms or had had some improvement when evaluated approximately 4 months after the treatment; 30% reported temporary improvement, and in 15% there was no change compared with the pre-treatment symptoms. In the recent study [28] only 15% of the failed back surgery patients got relief from their symptoms after a single CT- or MR-guided nerve root infiltration, and after four treatments 72% of the patients were symptom free.

In our study patients with irritation after failed back surgery had excellent outcome from nerve root infiltration therapy with almost 80% of initial pain free result. It has been reported that 75.4% of patients with radicular leg pain had a successful long-term outcome after lumbar transforaminal epidural steroid injection [3]. Twenty-two of the 28 patients with lumbar disc herniation had considerable and sustained relief from their symptoms in another study [4]. The outcome was not evaluated separately for lumbar and sacral nerve roots in these studies. In our series the outcome of nerve root infiltration therapy was in accordance with these other studies using fluoroscopy or CT guidance [3, 4, 10, 30]. Our previous controlled trial is in accordance with these studies, too [6].

In our study the patients with normal to minor disc pathology and patients with nerve root channel narrowing due bony or scar involvement did not have such a good outcome from nerve root infiltration therapy. This is probably due to more passive nature of the underlying disease. In these patient groups the feasibility of the procedure remains controversial. It must be stated, however, that percutaneous nerve root infiltration therapy is an almost complication-free and patient-friendly procedure when compared with surgery, can be very easily repeated, and can be mastered by any radiologist.

On the basis of the present study MRI-guided needle introduction is an accurate procedure and may be used to substitute conventional techniques for nerve root infiltration. This would be especially useful in cases where there are substantial anatomical or structural changes due variation or pathology. We conclude that MRI-guided nerve root infiltration therapy is safe, accurate, and easy to perform. Nerve root infiltration therapy is an effective treatment form to control radicular pain and can be used to substitute more invasive procedures in selected patient groups.

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